



Image Retrieval

Mid term project

Rensso Victor Hugo Mora Colque

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Motivation and Context

The exponential growth of digital image collections across a wide range of applications, such as social media, surveillance systems, and multimedia archives, has created the need for efficient methods for searching and retrieving visual information. In this context, the problem of *image retrieval* consists of, given a query image, automatically retrieving images from a database that are visually similar or that represent the same semantic content.

One of the most influential approaches in classical computer vision literature is the *Bag of Visual Words* (BoVW) model, which adapts concepts from natural language processing to the visual domain. In this approach, images are represented as histograms of visual words constructed from local descriptors, such as SIFT, which are previously quantized using clustering algorithms [1]. This paradigm enabled retrieval systems to scale efficiently to large image collections.

The use of invariant local descriptors, such as SIFT, has been fundamental in the development of robust image retrieval systems, as they allow handling variations in scale, rotation, and illumination [2]. Building on this foundation, several works have proposed improvements, including weighting schemes, spatial pyramids, and more sophisticated representations such as *Fisher Vectors*, which capture higher-order statistics of local descriptors [3].

In this project, the problem of *image retrieval* will be addressed using classical computer vision techniques, with emphasis on the construction of visual representations and the analysis of similarity metrics. Standard datasets such as Oxford and Paris Buildings will be used, which have been widely employed as benchmarks for evaluating image retrieval systems [4, 5].

Dataset Description

In this project, we make use of the Oxford and Paris Buildings datasets ([link](#)), which are widely recognized benchmarks for evaluating image retrieval systems. The version used in this work is available on Kaggle and consists of images collected from Flickr depicting well-known landmarks from the cities of Oxford and Paris.

The Oxford Buildings dataset contains 5,062 images, while the Paris Buildings dataset includes 6,412 images [4]. These images were gathered by querying Flickr using the names of specific landmarks, which results in a realistic and challenging dataset due to variations in viewpoint, scale, illumination, occlusions, and background clutter. A key characteristic of these datasets is the availability of manually annotated ground truth. In the case of the Oxford dataset, images are associated with 11 different landmarks, each defined through multiple query instances. This leads to a set of queries for which relevant images are labeled, allowing a precise evaluation of retrieval performance. Similarly, the Paris dataset follows the same structure and evaluation protocol. Additionally, the datasets may include distractor images that do not belong to any landmark category, further increasing the difficulty of the retrieval task. This setup reflects real-world scenarios, where irrelevant images are commonly present in large-scale databases.

Evaluation Protocol

To objectively evaluate the performance of the proposed image retrieval systems, a standard retrieval protocol will be followed. Given a query image, the system must return a ranked list of images from the database ordered by decreasing similarity.

Query and Ranking

For each query image, the system computes a similarity score between the query and all images in the dataset. The output of the system is a ranking of images sorted according to this similarity measure.

Each query is associated with a set of *relevant images*, defined by the ground truth annotations provided in the dataset. These relevant images correspond to the same landmark as the query.

Evaluation Metrics

The performance of the system will be evaluated using the following metrics:

Precision@k Precision at rank k (Precision@k) measures the proportion of relevant images among the top- k retrieved results:

$$\text{Precision@k} = \frac{\text{\#relevant images in top } k}{k} \quad (1)$$

This metric evaluates how many of the top retrieved images are correct.

Average Precision (AP) Average Precision summarizes the precision-recall curve for a single query:

$$\text{AP} = \frac{1}{N} \sum_{k=1}^M P(k) \cdot \text{rel}(k) \quad (2)$$

where $P(k)$ is the precision at rank k , $\text{rel}(k)$ is an indicator function that is 1 if the item at rank k is relevant and 0 otherwise, M is the total number of retrieved images, and N is the total number of relevant images for that query.

Mean Average Precision (mAP) The overall performance across all queries is measured using Mean Average Precision (mAP):

$$\text{mAP} = \frac{1}{Q} \sum_{i=1}^Q \text{AP}_i \quad (3)$$

where Q is the number of queries and AP_i is the Average Precision for the i -th query.

Qualitative Evaluation

In addition to quantitative metrics, students must include a qualitative analysis of their system. For a subset of query images, the top-5 retrieved results must be visualized and discussed. Students should analyze both successful and failure cases, identifying possible reasons related to the chosen feature representation and similarity measure.

Project Task

The objective of this project is to design and implement an *image retrieval system* using classical computer vision techniques. Given a query image, the system must retrieve and rank images from a database according to their visual similarity.

General Description

Students are required to build a complete image retrieval pipeline that takes as input a query image and returns a ranked list of images from the dataset. The system must rely exclusively on classical feature extraction and similarity computation techniques, without the use of deep learning models.

The dataset to be used is the Oxford and/or Paris Buildings dataset. Each image in the dataset belongs to a particular landmark, and the goal is to retrieve images corresponding to the same landmark as the query.

System Requirements

The implemented system must include the following components:

1. Feature Extraction Students must extract visual features from all images in the dataset. At least two different types of feature representations must be implemented and compared:

- **Local feature-based representation:** e.g., SIFT descriptors.
- **Global or structured representation:** e.g., HOG descriptors or alternative global features.

Additionally, students are encouraged to explore more advanced representations such as Bag of Visual Words (BoVW) or Fisher Vectors.

2. Visual Vocabulary (if applicable) If using a Bag of Visual Words approach, students must:

- Construct a visual vocabulary using a clustering algorithm (e.g., k-means).
- Represent each image as a histogram of visual words.

The size of the vocabulary must be justified experimentally.

3. Image Representation Each image must be encoded as a fixed-length feature vector. If multiple feature types are used, students may:

- Evaluate them independently, or
- Combine them into a single representation (feature concatenation).
- You can propose your own model or descriptor

4. Similarity Measure and Retrieval Students must define and implement a similarity (or distance) function to compare images. Possible choices include:

- Euclidean distance
- Cosine similarity
- Other justified metrics

Given a query image, the system must:

- Compute its feature representation
- Compare it against all images in the dataset, you can employ data structures to speed up the retrieval model
- Return a ranked list sorted by similarity

5. Retrieval Output For each query, the system must return at least the top-5 most similar images. These results must be used for both quantitative and qualitative evaluation.

Experimental Requirements

Students must conduct a systematic evaluation of their approach, including:

- Comparison of at least two different feature representations
- Analysis of different similarity metrics (if applicable)
- Study of the effect of key parameters (e.g., vocabulary size in BoVW)

Restrictions

To ensure a fair evaluation and focus on classical computer vision techniques, the following restrictions apply:

- The use of deep learning models (e.g., CNNs, Vision Transformers, CLIP) is **not allowed**
- Pretrained feature extractors based on neural networks are **not allowed**
- The system must be implemented using classical methods, you can employ other different models presented at class, however you must deeply understand them

Expected Outcomes

At the end of the project, students should be able to:

- Understand and implement a complete image retrieval pipeline
- Analyze the strengths and limitations of different visual representations
- Evaluate retrieval systems using standard metrics such as Precision@k and mAP
- Interpret qualitative results and identify failure cases

Grading Criteria

The final grade of the project will be computed based on four components: report quality, presentation, individual understanding, and system performance.

Grade Distribution

The grading is distributed as follows:

- **Report (4 points):** clarity, structure, methodology, experimental analysis, and discussion of results.
- **Presentation (5 points):** clarity of explanation, organization, and ability to communicate ideas effectively.
- **Individual Understanding (3 points):** evaluation of each student's understanding of the implemented system, including questions during the presentation.
- **Model Performance (8 points):** objective evaluation of the image retrieval system.

Performance Evaluation

The performance score is based on the ranking quality of the retrieval system. In particular, the Mean Average Precision (mAP) will be used as the primary evaluation metric.

Let mAP_i be the performance of group i , and let:

- mAP_{best} be the highest mAP achieved among all groups
- mAP_{worst} be the lowest mAP achieved among all groups

The performance score for each group is computed as a normalized score:

$$G_{performance}^{(i)} = 8 \cdot \frac{mAP_i - mAP_{worst}}{mAP_{best} - mAP_{worst}} \quad (4)$$

This normalization ensures that:

- The best-performing group receives the maximum score (8 points)
- The lowest-performing group receives the minimum score (0 points)
- Other groups are graded proportionally

Robustness Considerations

To ensure fairness:

- All models must be evaluated using the same dataset and query set.
- Students must provide their evaluation code to allow reproducibility.
- Any inconsistencies or errors in evaluation may result in penalties.

Qualitative Assessment

In addition to quantitative performance, students must provide a qualitative analysis of their results. This includes:

- Visualization of top retrieved images for selected queries
- Identification of failure cases
- Discussion of limitations of their approach

References

- [1] G. Csurka, C. Dance, L. Fan, J. Willamowski, and C. Bray, *Visual categorization with bags of keypoints*, Workshop on Statistical Learning in Computer Vision, ECCV, 2004.
- [2] D. G. Lowe, *Distinctive image features from scale-invariant keypoints*, International Journal of Computer Vision, vol. 60, no. 2, pp. 91–110, 2004.
- [3] F. Perronnin, J. Sánchez, and T. Mensink, *Improving the Fisher Kernel for Large-Scale Image Classification*, European Conference on Computer Vision (ECCV), pp. 143–156, 2010.
- [4] J. Philbin, O. Chum, M. Isard, J. Sivic, and A. Zisserman, *Object retrieval with large vocabularies and fast spatial matching*, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2007.
- [5] J. Philbin, O. Chum, M. Isard, J. Sivic, and A. Zisserman, *Lost in quantization: Improving particular object retrieval in large scale image databases*, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2008.